

Descriptive Vs Inferential

	Descriptive	Inferential
Purpose	Summarize & describe data	Make predictions & draw conclusions from data
Works with	Actual data you collected	Goes beyond the data to generalize to a population
Examples	Mean, median, mode, Range, Standard deviation, Graphs	Hypothesis Testing, Confidence Intervals, P-value, t-tests
Focus	"What does the data say?"	What can we infer about the population

→ for example :- Let's say you survey 100 students on their exam scores.

→ Descriptive Stats :-

"the avg score is 72.3, & the Standard deviation is 8.5"

→ you are just describing what you observed.

→ Inferential Stats :-

"We are 95% Confident that the avg score of all students in the school is b/w 70 & 75."

→ you are inferring something about a larger population based on your sample.

→ we have already covered descriptive stats, now its time to start with inferential stats.

→ It's the branch of statistics that helps us make predictions or decision about a population based on sample of data.

